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AtrilMatic: A Web-based Smart Music Stand

Por/By???
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Acknowledgments

Viva el director

Abstract

AtrilMatic: A Web-based Music-Stand

Keywords: Digital music stand, pitch detection, score following, real-time audio processing, MusicXML, WebAssembly, Rust.

Resumen

AtrilMatic, un atril inteligente

Palabras clave: Atril digital, detección de tono, seguimiento de partituras, procesamiento de audio en tiempo real, MusicXML, WebAssembly, Rust.

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Chapter 1. Introduction

1.1. Motivation

1.2. Objectives / goals ??

1.2.1. General objective

1.2.2. Specific objectives

1.3. Planning

Chapter 2. Related work

2.1. Introduction

2.2. Technological foundations

2.3. Existing solutions

Before starting with our own app, we felt that a way of gathering... was looking into already existing solutions in the market to see what we liked or not...???

This section reviews 4 applications that propose similar solutions to what we wanted to do. The solutions analysed are: Tomplay, Oktav, Newzik and Enote.

2.3.1. Tomplay

Tomplay is a mobile and desktop app developed by Tomplay.

maybe we can add here screenshot of the UI from the original web

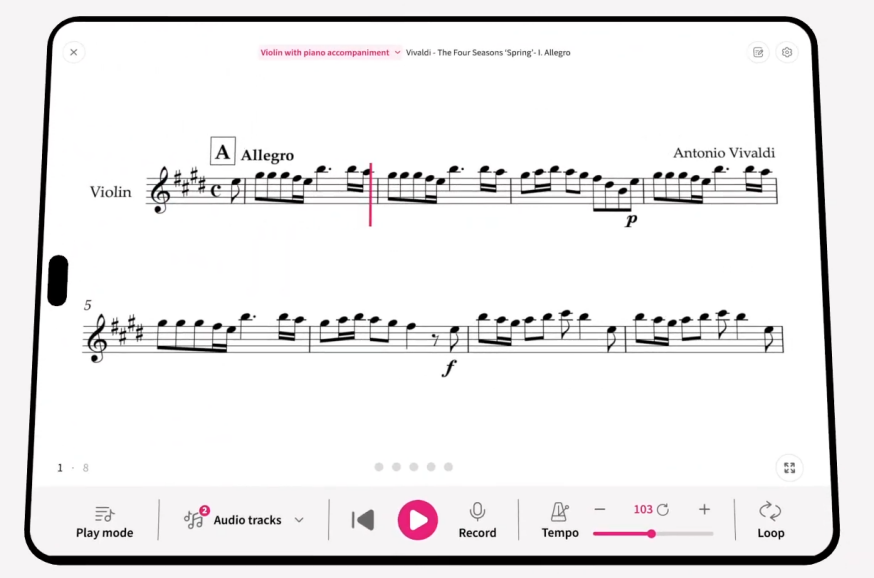


Figure 2.1: Tomplay user interface.

Its main features are:

- **Digital sheet music**: Automatically scrolls in sync with the music.
- **Note by note practice**: The score advances only when you play the right note.
- **Adjust the tempo**: Slow down or speed up the music tempo.
- **Record yourself**: Record and listen back to your performance.
- **Annotate**: Add your own annotations to the music sheet.

- **Big music sheet catalogue**: Over 100,000 scores with professional audio accompaniments.

However, it presents some limitations:

- Paid subscription
-
-

2.3.2. OKTAV

OKTAV is a web app developed by OKTAV.



Figure 2.2: Oktav user interface.

Its main features are:

- **Autoscroll**: Play without the need to turn pages and adjust tempo.
- **Annotate**: Ability to add fingerings, notes or highlight sections.
- **Play-along**: listen and play along to sheet music based on MIDI.
- **Digital piano**: See how to play a music sheet on a visual keyboard shown along the music sheet.

However, it presents some limitations:

- Paid subscription
- Only for piano, violin and cello
-

2.4. Comparison with the proposed solution

Chapter 3. Development

3.0.1. Functional requirements

3.0.2. Non-functional requirements

3.1. Design

3.1.1. Technical Challenges

It is important to commence by outlining the actual technological needs an application like Atril would require. Recognizing pitch dynamically from microphone input could, in general, be considered a resource-intensive and in some ways non-deterministic task for a computer to perform. It is clear that a system made to perform such a task should be capable of adjusting to different microphones as well as different noise levels, and of recognizing frequencies with a minimal margin of error. The fact that, in the case of this application, such a task would have to be performed dynamically and in real time would mean that it would prove to be even more challenging. Furthermore, there is a prediction or matching element that uses the pitch data produced by the recognition system to give the best possible estimate of the position of the player in accordance with the musical composition being played. It is important to highlight that this is probably where the biggest technological focus should be allocated when it comes to an application like Atril. The recognition and prediction logic will run multiple times a second while also having to be lightweight and easy to implement into the vast world of web development frameworks, languages, and established principles.

3.1.2. WebAssembly

This is precisely where WebAssembly comes in. WebAssembly is a bytecode that acts as a compilation target for many, usually low-level, programming languages, famously C, C++, Rust, and Go. It offers simple validation, fast compilation, and interoperability with the web platform, including direct access to JavaScript and Web APIs [Haas et al., 2017].

3.1.3. Why WebAssembly?

Although WebAssembly was initially widely adopted mostly as a platform for cryptomining, this has since become a marginalized minority, comprising less than 1% of the binaries found on the web among the 8,461 WebAssembly binaries studied by the University of Stuttgart in their 2021 paper *An Empirical Study of Real-World WebAssembly Binaries* [Hilbig et al., 2021], showing a wide adoption of the format for a variety of use cases.

WebAssembly as a compilation target for the note recognition and prediction system outlined earlier is a clear choice due to several important factors. Firstly, it offers a low-overhead means of execution that will not bog down or make the rest of the proposed application heavier. Additionally, it offers a compilation target for languages that are significantly faster than JavaScript, allowing the creation of a performant library that

is still entirely interoperable with the web world and its systems via a simple API system. Finally, and importantly, WebAssembly is an abstraction over modern hardware, meaning that our application will still remain platform independent.

3.1.4. Limitations and Solutions

It is important to note the limitations of WebAssembly and the measures that can be taken to minimize their impact. Hilbig et al. [Hilbig et al., 2021], in their study of 8,461 real-world binaries, found that "vulnerabilities that propagate from insecure source languages potentially affect a wide range of binaries (e.g., two thirds of the binaries are compiled from memory unsafe languages, such as C and C++)". This matters because WebAssembly's built-in safeguards do not neutralize such bugs. The choice of source language, and its safety, is therefore crucial when developing a WebAssembly system.

3.1.5. Rust and WebAssembly

Among the languages commonly compiled to WebAssembly, Rust is the strongest candidate with respect to memory safety. Rust's compiler enforces ownership and borrowing: a value has one owner, and there can be either one mutable reference or many shared ones, never both. For safe Rust, this rules out data races and invalid memory accesses, a property that has been formally proven in machine-checked form for a realistic subset of the language and for several core standard-library abstractions [Jung et al., 2018]. Importantly, this is achieved without garbage collection, whereas languages such as Go rely on runtime memory checking and garbage collection [Xu et al., 2021]. Rust's guarantees therefore come largely from compile-time analysis rather than runtime overhead, which is essential for a real-time audio application.

3.1.6. WebAssembly Compilation Process

What follows is a small explanation of the WebAssembly compilation process, sufficient to understand the functionality of our system. In our case, Rust uses the LLVM intermediate representation, allowing a single intermediate representation to be compiled to a variety of different targets. In this case, LLVM has a WebAssembly target that converts the intermediate representation given by the Low-Level Virtual Machine into a `.wasm` module, a compact, stack-based bytecode in which high-level Rust constructs such as structs and slices are lowered to a small set of primitive numeric types and offsets into a linear memory, and in which control flow is structured [Haas et al., 2017]. Because WebAssembly can only exchange numbers with JavaScript directly, a tool such as `wasm-bindgen` post-processes the module and generates the JavaScript and TypeScript glue code that translates richer types such as strings and objects across the boundary [Rust and WebAssembly Working Group, 2026]. At load time, the browser engine validates the module against WebAssembly's type system before executing anything, which is what allows untrusted code to be run safely [Haas et al., 2017]. The format was designed to be fast to validate and compile, and the engine then translates it to machine code and runs it inside a sandbox that confines it to its own linear memory [Haas et al., 2017].

3.1.7. System architecture overview

3.1.8. Database design – we dont have DB!

3.2. Implementation

3.2.1. Backend and persistence

3.3. Key Algorithms

(Temporary) This section is a placeholder and will be expanded.

At the heart of detecting a pitch from time-domain data stands the Fast Fourier Transform (FFT). In this project, the FFT is provided by the `rustfft` crate [Mahler, 2026] and is combined with several self-implemented window functions, which are discussed further below. The FFT converts the time-domain data obtained from the microphone input into frequency-domain data, which is then used by the pitch detection algorithms, although some of them, such as the YIN algorithm, still work directly with time-domain data.

3.3.1. Fast Fourier Transform Algorithm

The Fourier transform is a mathematical operation that converts a signal, given as an amplitude through time, into a spectrum in which each frequency present in the signal is displayed together with its amplitude. It rests on the Fourier series, which states that a periodic signal $f(t)$ with period T can be written as a sum of sine and cosine waves whose frequencies are integer multiples of the fundamental frequency $\omega_0 = 2\pi/T$:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right]. \tag{3.1}$$

Essentially, the Fourier transform shows which frequencies make up a given signal and in which proportions. In effect, it determines how much of each sine and cosine wave is needed to reconstruct the signal. To find how much of frequency F a signal contains, the signal is multiplied by a wave of frequency F and the average is taken. If the signal contains that frequency, the product is mostly positive and the average is large. Otherwise, the positive and negative parts cancel and the average is near zero. Using a cosine copy and a sine copy makes the answer independent of the phase of the wave. For the series above, this is exactly how the coefficients are obtained:

$$a_n = \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t)\, dt, \qquad b_n = \frac{2}{T} \int_0^T f(t) \sin(n\omega_0 t)\, dt. \tag{3.2}$$

Repeating this for every frequency yields the spectrum.

In practice, the microphone delivers a finite number N of discrete samples $x_0, \ldots, x_{N-1}$, so the discrete Fourier transform (DFT) is used. With the principal Nth root of unity W it can be written compactly as [Cooley and Tukey, 1965]

$$X_k = \sum_{n=0}^{N-1} x_n W^{kn}, \qquad k = 0, 1, \ldots, N-1, \qquad W = e^{-2\pi i/N}. \tag{3.3}$$

By Euler's formula, $W^{kn} = \cos(2\pi kn/N) - i\sin(2\pi kn/N)$, so Equation 3.3 multiplies the signal by a cosine copy and a sine copy and sums the result, which is the discrete counterpart of the averaging described above. Bin k corresponds to the frequency $f_k = kf_s/N$, where f_s is the sampling rate, and the magnitude $|X_k|$ gives the strength of that frequency in the signal.

Evaluating Equation 3.3 directly requires on the order of N^2 operations. The FFT is not a different transform but a fast way of computing this one: the algorithm of Cooley and Tukey needs, in their words, "a number of operations proportional to $N \log N$ rather than N^2" [Cooley and Tukey, 1965]. This difference is what makes it feasible to run the analysis several times per second on live microphone input. In the current implementation, the sample rate is 48 kHz, the FFT and window sizes are 2048 samples, and the hop size is 512 samples, so a new spectrum is computed every 512 samples, approximately 94 times per second.

Expand with the exact FFT implementation details.

3.3.2. Window Functions

When computing the FFT, the algorithm behaves as if it were being applied to a spectrum that repeats infinitely. A wave that is cut off at the edge of the block therefore creates a noticeable elevation in amplitude which skews the results. This phenomenon is called spectral leakage. To reduce it, different window functions are applied to the signal before the FFT is computed. The general idea behind all window functions is to fade the edges of the block, although this is achieved through different means. The error can be reduced by multiplying the block with a window function $w[n]$ that fades the signal towards zero at both ends before the transform is computed:

$$y[n] = w[n]\,x[n], \qquad n = 0, 1, \ldots, N-1. \tag{3.4}$$

Every window is a compromise between two properties of its spectrum: the width of the main lobe, which determines how close two frequencies can be before they blur into one, and the height and decay of the side lobes, which determine how much a strong component leaks into its surroundings [Harris, 1978]. The four windows used in this project all belong to the family of cosine-sum windows,

$$w[n] = \sum_{m=0}^{M} (-1)^m a_m \cos\left(\frac{2\pi mn}{N-1}\right), \qquad n = 0, 1, \ldots, N-1, \tag{3.5}$$

and differ only in the number of terms and in the coefficients a_m.

Rectangular Window

The rectangular window does not modify the signal at all ($M = 0$, $a_0 = 1$):

$$w_{\mathrm{R}}[n] = 1. \tag{3.6}$$

It has the narrowest main lobe of the four, so it separates close frequencies best, but its side lobes are the highest and decay slowly. It is only a good choice when the signal is guaranteed to be periodic within the block.

Hann Window

The Hann window is a single raised cosine ($M = 1$, $a_0 = a_1 = 0.5$) that reaches exactly zero at both ends of the block:

$$w_{\text{Hann}}[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{N-1}\right). \tag{3.7}$$

Because the block ends at zero, the repeated copies join smoothly, and the side lobes are low and fall off quickly. The price is a main lobe twice as wide as that of the rectangular window.

Hamming Window

The Hamming window uses the same shape with different coefficients ($a_0 = 0.54$, $a_1 = 0.46$):

$$w_{\text{Hamming}}[n] = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right). \tag{3.8}$$

These are the commonly used rounded values of the coefficients. The window does not quite reach zero at the ends, which lowers the nearest side lobes further than the Hann window does, but a small discontinuity remains, so the distant side lobes decay slowly.

Blackman Window

The Blackman window adds a second cosine term ($M = 2$, $a_0 = 0.42$, $a_1 = 0.5$, $a_2 = 0.08$):

$$w_{\text{Blackman}}[n] = 0.42 - 0.5\cos\left(\frac{2\pi n}{N-1}\right) + 0.08\cos\left(\frac{4\pi n}{N-1}\right). \tag{3.9}$$

It gives the lowest leakage of the four and its side lobes fall off quickly, at the cost of the widest main lobe. It is useful when a weak component has to be found next to a much stronger one.

Comparison

Table 3.1 summarizes the trade-off. The main lobe width is measured in FFT bins, where one bin corresponds to a frequency step of f_s/N. The coherent gain is the mean value of the window, $\frac{1}{N}\sum_{n=0}^{N-1} w[n]$; since the windowed signal is scaled by this factor, amplitudes read from the spectrum must be divided by it.

Window	Main lobe width (bins)	Highest side lobe (dB)	Side-lobe decay (dB/octave)	Coherent gain
Rectangular	2	−13	6	1.00
Hann	4	−31	18	0.50
Hamming	4	−43	6	0.54
Blackman	6	−58	18	0.42

Table 3.1: Approximate properties of the four windows. The main lobe width is the distance between the first nulls on both sides of the peak.

Chapter 4. Evaluation

4.0.1. Objectives / goals ??

4.0.2. Participants

4.0.3. Test environment

4.0.4. Evaluation tasks

4.0.5. Procedure

Chapter 5. Conclusions

5.1. Conclusions

5.1.1. Fulfilment of the objectives / goals ??

Capítulo 1. Introducción

1.1. Motivación

1.1.1. Objetivo general

1.1.2. Objetivos específicos

Capítulo 2. Conclusiones

2.1. Conclusiones

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AppendixA
Evaluation questionnaire

This appendix presents ... Eg: SUS survey